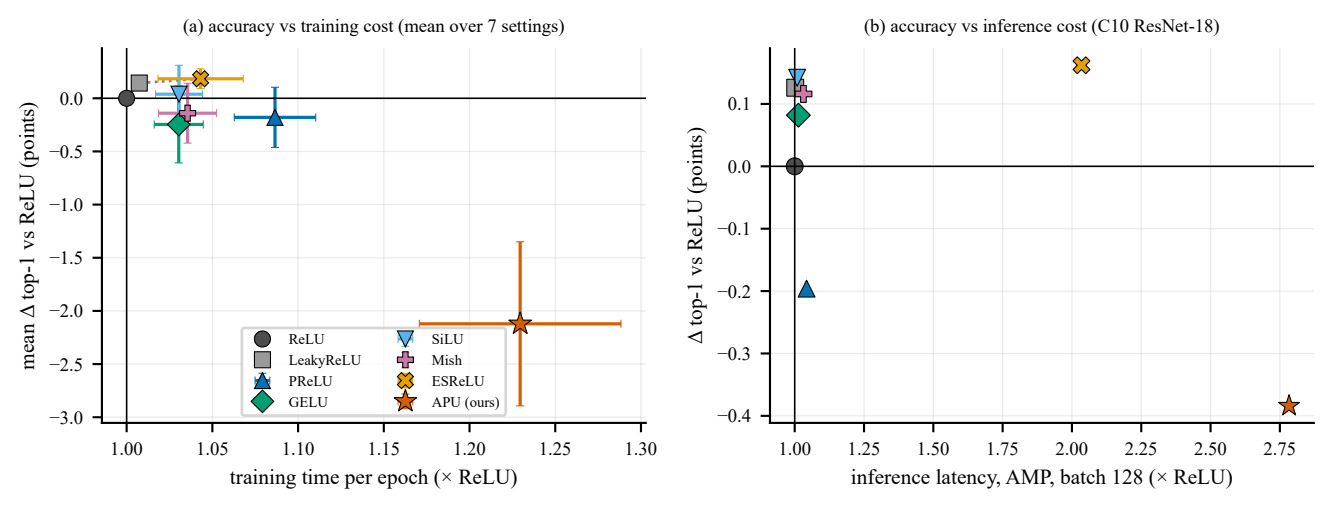
(a) accuracy vs training cost (mean over 7 settings)
mean Δ top-1 vs ReLU (points)
training time per epoch (× ReLU)
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
1.00
1.05
1.10
1.15
1.20
1.25
1.30
ReLU
LeakyReLU
PReLU
GELU
SiLU
Mish
ESReLU
APU (ours)
(b) accuracy vs inference cost (C10 ResNet-18)
Δ top-1 vs ReLU (points)
inference latency, AMP, batch 128 (× ReLU)
0.1
0.0
−0.1
−0.2
−0.3
−0.4
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75